Sim2Real-HAD: A Simulation-Driven End-to-End Hyperspectral Anomaly Detection Framework

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***Abstract*—To address the lack of end-to-end learning capability and the scarcity of real-world data in existing hyperspectral anomaly detection (HAD) methods, which limit model performance, this article proposes a simulation-driven end-to-end HAD framework. First, a background–anomaly simulation pipeline tailored for anomaly detection (AD) is developed to automatically generate hyperspectral images with diverse backgrounds and complex distribution characteristics, providing scalable large-scale training data for model learning. Second, an end-to-end HAD network is designed based on the simulated data, enabling a direct mapping from raw hyperspectral observations to pixel-level AD results, thereby jointly optimizing feature extraction and anomaly discrimination. Furthermore, Reed–Xiaoli (RX) residual constraints, anomaly sparsity constraints, and spectral similarity constraints (SSCs) are embedded into the training process to enhance anomaly discrimination capability and cross-scene generalization performance. Extensive experiments conducted on 113 public hyperspectral scenes demonstrate the superior detection accuracy and generalization ability of the proposed method. The code will be available at https://github.com/fomanwong/Sim2Real**

***Index Terms*—Anomaly detection (AD), deep learning, hyperspectral imagery (HSI), unified model.**

I. Introduction

HYPERSPECTRAL imagery (HSI), characterized by its capability of jointly capturing spatial and spectral information, has been widely applied in the field of remote sensing [1], [2]. As a critical task in hyperspectral image analysis, hyperspectral anomaly detection (HAD) [3], [4], [5], [6] aims to identify rare pixels whose spectral signatures deviate from the surrounding background in complex scenes, and has found important applications in precision agriculture, environmental monitoring, and national defense.

The challenge of HAD lies in its unsupervised nature, where anomaly targets are unknown during training and lack explicit semantic labels or prior category information. Traditional HAD methods can generally be divided into statistical model-based and representation-based approaches. Statistical methods are typically represented by the classical Reed–Xiaoli (RX) anomaly detector [7], which models background pixels as high-probability samples in a statistical distribution and anomalies as low-probability outliers. Based on this principle, various extensions, such as local RX [8], [9] and kernel RX [10], [11], have been proposed to improve adaptability to complex backgrounds. Since the Gaussian assumption of RX often fails in real scenes, background enhancement techniques [12], [13], [14], [15] were further introduced to enhance background–anomaly separability. Representation-based methods mainly include sparse representation and collaborative representation approaches. Sparse representation methods [16], [17], [18], [19] exploit the low-rank property of backgrounds and the sparsity of anomalies for anomaly separation, while collaborative representation methods [20], [21], [22], [23] assume that backgrounds can be effectively reconstructed from neighboring pixels whereas anomalies cannot. Although these methods improve detection performance to some extent, their assumptions often deviate from complex real-world scenarios, resulting in limited detection accuracy and poor generalization capability.

With the development of deep learning, data-driven HAD methods [24], [25], [26], [27], [28], [29] have become mainstream. Existing methods can be broadly divided into two paradigms based on their generalization ability. The first paradigm consists of scene-dependent background reconstruction-based methods, which typically employ autoencoders [29], [30] or generative adversarial networks [31], [32] to reconstruct background information and measure anomaly scores based on reconstruction errors. Despite achieving certain progress, these methods suffer from inherent limitations. First, their effectiveness relies on the assumption that background is easy to reconstruct, while anomalies are difficult to reconstruct. In complex hyperspectral scenes with highly nonuniform background structures, this assumption may not always hold, leading to background reconstruction errors and unintended reconstruction of anomalies. Second, these methods are usually trained and tested on a single scene, learning scene-specific static background distributions rather than general anomaly–background discrimination rules, which severely restricts cross-scene generalization and practical applicability.

To overcome these limitations, the second paradigm aims to construct generalizable "train-once" models. Representative methods include the anomaly-enhanced transformation network (AETNet) proposed by Li et al. [33] and the feature enhancement network (FEN) proposed by Jin et al. [34], which reconstruct the background during training to enhance its consistency, enabling spectral reconstruction and RX detection to be directly performed during inference without additional parameter adjustment for background reconstruction.

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Although AETNet and FEN improve detection performance, they still require a large amount of real-world background data for training and involve a two-step inference process of background reconstruction followed by RX detection. More recently, Li et al. [35] proposed the first end-to-end anomaly detection (AD) model, transferred direct detection (TDD), which reduces dependence on real-world data by simulating local-scale anomalies. This model can be trained on a single real-world scene; however, inference requires patch-wise processing followed by stitching.

In summary, the scarcity of real-world data limits the performance improvement of "train-once" models. Although TDD alleviates the data insufficiency issue to some extent through local patch-based training, its localized strategy fails to effectively model global scene context and requires target-dependent image resizing during testing, which may restrict its applicability in complex backgrounds and cross-scene scenarios and significantly reduce its practical usability. In addition, existing HAD networks generally lack effective prior constraints and generalization-enhancement mechanisms, making them sensitive to background variations and resulting in limited generalization capability.

To address these challenges, this article proposes a simulation-driven end-to-end HAD framework. First, to overcome the limitation of scarce real-world data, a background–anomaly simulation pipeline for HAD (HADSim) is developed to programmatically generate hyperspectral images with diverse backgrounds and distributions, providing high-quality training samples for end-to-end learning. Second, an end-to-end HAD network (Sim2Real) is designed based on the simulated data, enabling direct mapping from the entire input hyperspectral image to AD results, thereby improving detection performance and practicality. Finally, three key constraints (3C) are introduced: a residual constraint with RX statistical priors (RXC) and a sparse spatial distribution constraint for anomalies (ASC) are imposed on the network output to enhance detection performance, while a spectral similarity constraint (SSC) is applied to the encoder to encourage learning of spectral deviation patterns between anomalies and background. These constraints jointly improve anomaly discrimination capability and cross-scene generalization. The main contributions of this article are summarized as follows.

1) A simulation-driven training paradigm is proposed, along with a background–anomaly simulation pipeline for HAD, effectively alleviating the limitation of scarce real-world data and enabling end-to-end HAD model training.
2) An end-to-end HAD network based on simulated data is designed to directly map entire input hyperspectral images to AD results, improving detection performance and practicality.
3) RX residual constraints, anomaly sparsity constraints, and SSCs are introduced to guide the model learning under multiple priors, enhancing anomaly discrimination capability and generalization performance.
4) Extensive experiments demonstrate that the proposed Sim2Real-HAD achieves superior detection accuracy and generalization capability compared with existing mainstream HAD methods.

The rest of this article is organized as follows. Section II introduces deep learning-based HAD and contrastive learning methods. Section III describes the overall framework and key components of Sim2Real-HAD. Section IV presents comparative experimental results and model analysis. Section V concludes the article and discusses future work.

II. Related Work

The proposed Sim2Real-HAD model performs anomaly segmentation using a deep network with SSC under a contrastive learning paradigm. This section briefly reviews deep AD methods, followed by contrastive learning-based semantic segmentation approaches.

A. Deep-Learning-Based AD

Deep learning-based HAD methods have been extensively studied [36], [37], [38]. Most existing approaches are based on the assumption that background pixels are easier to reconstruct than anomaly pixels. These methods reconstruct background information through deep networks and detect anomalies according to the discrepancy between reconstructed and original images. Their main objective is to accurately reconstruct backgrounds while suppressing or enlarging anomaly reconstruction errors. For example, Wang et al. [39] proposed Auto-AD, which employs a fully convolutional autoencoder with skip connections and an adaptive weighted loss to suppress anomaly reconstruction. Su et al. [40] introduced DeepSeT, which incorporates a binary mask into the loss function to shield potential anomaly pixels and improve background reconstruction accuracy.

To further enhance reconstruction quality, many studies introduce sparse or low-rank priors based on prior assumptions about background and anomaly distributions. Wang et al. [41] proposed DeepLR, which incorporates a low-rank prior during optimization to reconstruct dominant background components while suppressing sparse anomalies. Lian et al. [42] proposed LRMAE, which imposes an implicit low-rank constraint by restricting encoder feature channels and further enhances anomaly separation through sparsification in the loss stage.

In addition, some methods incorporate clustering strategies to guide background reconstruction and reduce anomaly interference. Song et al. [43] proposed GSGCRD, which employs K-means clustering to filter small-sample clusters and improve reconstruction stability. Li et al. [44] proposed a GAN-based framework that uses DBSCAN clustering to identify background pixels for sparse reconstruction, thereby improving background–anomaly separability.

Although the above methods achieve good AD performance, they suffer from low efficiency, as they require retraining the background reconstruction model for each test scene. To mitigate this issue, some studies [33], [34] attempt to train networks using large amounts of real-world background samples, but data scarcity still hinders further performance improvement. Meanwhile, most existing methods adopt a two-stage pipeline–background reconstruction (or enhancement)

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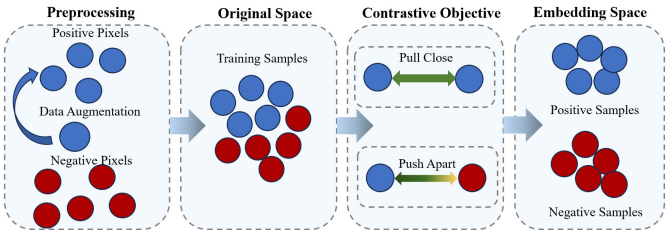


Fig. 1. Pixel-wise contrastive learning mechanism.

followed by AD. In contrast, the TDD method [35] is the first to propose an end-to-end AD framework and alleviates data scarcity through an image patching strategy. However, patch-based image processing strategies limit the model's ability to extract global contextual information and reduce its flexibility in practical applications. Moreover, these methods lack effective prior constraints, which limits their generalization capability and detection accuracy. Therefore, it is necessary to develop an end-to-end network that maps the entire input hyperspectral image to AD results, along with effective constraints to improve detection performance and practicality.

B. Contrastive-Learning-Based Semantic Segmentation

Contrastive learning, as a powerful self-supervised representation learning paradigm, aims to uncover the intrinsic structure of data without relying on manual annotations by constructing positive and negative sample pairs, thereby obtaining highly discriminative feature representations.

As shown in Fig. 1, the core idea of contrastive learning is to pull semantically similar pairs (positive pairs) closer in the feature space while pushing apart semantically dissimilar pairs (negative pairs). Based on this idea, Ji et al. [45] proposed IIC, which achieves image semantic segmentation by maximizing the mutual information between augmented views of the same image (positive pairs). Subsequently, Cho et al. [46] introduced PiCIE, which further incorporates geometric consistency constraints on top of IIC to enhance pixel-level clustering performance. Hamilton et al. [47] proposed STEGO, which leverages features extracted from pretrained vision models and reinforces clustering features through contrastive learning, achieving high segmentation accuracy in unsupervised semantic segmentation [48]. Seong et al. [49] introduced HP, a contrastive learning method that combines global and local implicit positive-pair construction strategies to effectively improve semantic consistency and inter-class separability. In addition, methods such as DepthG [50], EAGLE [51], and ReCLIP [52] further enhance the feature representation ability of semantic segmentation models by introducing contrastive learning strategies.

Although contrastive learning can obtain highly discriminative feature representations under unsupervised settings, the primary objective of HAD is to identify unknown targets that occupy only a few pixels. Therefore, due to the fundamental differences in data characteristics and task objectives, directly transferring contrastive learning to HAD presents significant challenges. To address this issue, this article innovatively introduces contrastive learning into the HAD model and

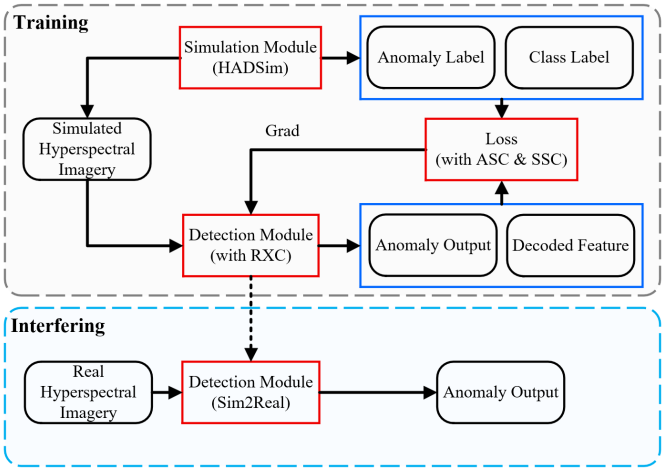


Fig. 2. Framework of the Sim2Real-HAD.

implements it as a SSC via an auxiliary decoder. The core idea is to use contrastive learning on the auxiliary decoder features to constrain and guide the encoder's representation learning process. Specifically, background and anomaly class information is recorded during data simulation, and the pixel-level feature similarities of the decoder are computed during training. If two pixels belong to the same class, their similarity is constrained to be 1; if they belong to different classes, their similarity is constrained to be 0, thereby encouraging the model to learn spectral similarity relationships. This strategy encourages the model to avoid relying on scene-specific superficial features for anomaly discrimination and instead focus on reasoning about anomalies based on spectral similarity relationships, thus learning more universal AD features. Therefore, the proposed SSC aims to leverage the strong feature disentangling ability of contrastive learning to force the encoder to learn scene-independent and physically consistent spectral representations. This strategy enables the model to acquire domain-invariant features that generalize to unseen scenes, thereby improving the generalization capability of HAD models.

III. Methodology

A. Sim2Real-HAD Framework

This article investigates HAD and proposes a simulation-driven end-to-end HAD framework, termed Sim2Real-HAD, whose overall pipeline is illustrated in Fig. 2. The Sim2Real-HAD framework consists of two stages: training and inference. During the training stage, a data simulation module is employed to generate hyperspectral images along with corresponding anomaly and land-cover class labels. These simulated data are then fed into an end-to-end HAD network with 3C for joint learning. Guided by the RXC prior information, the network produces anomaly responses and decoded features, while a joint loss function constructed from the associated labels—comprising ASC and SSC—is used to iteratively optimize the model parameters, thereby completing model training under the 3C constraints. In the inference stage, the trained model is directly applied to real hyperspectral

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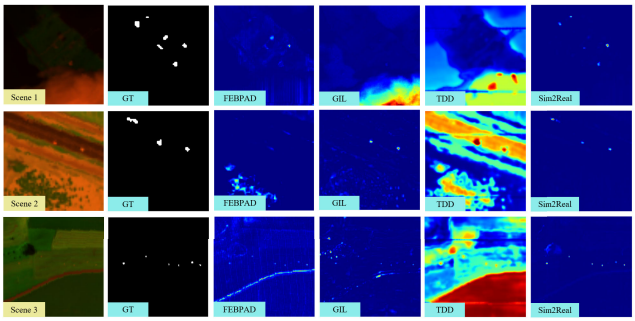


Fig. 3. AD results obtained by FEBPAD [53], GIL [54], TDD [35], and our proposed Sim2Real.

data for end-to-end AD, enabling a smooth transition from simulation to real-world scenarios.

Existing HAD methods that follow a "one-step training" paradigm generally lack end-to-end learning capability and suffer from data scarcity, limited practicality, and insufficient generalization ability, which ultimately degrades HAD performance. To address these issues, this article introduces three key innovations. First, a simulation strategy termed HADSim is proposed to jointly model backgrounds, anomalies, and noise. By simulating complex conditions such as background textures, spectral variability, and false anomalies, HADSim generates training data with rich contextual information and provides reliable priors for model learning. Second, an end-to-end HAD network is designed to improve practicality and generalization. Trained solely on simulated data, the proposed network directly produces anomaly maps from input hyperspectral images without additional postprocessing during inference. Third, three dedicated constraints are introduced to further enhance detection performance: 1) RXC incorporates RX detection results as prior guidance, enabling the network to refine anomaly responses by combining traditional priors with data-driven learning; 2) ASC imposes a scale-aware sparsity constraint on the anomaly response map to suppress large-scale false alarms while preserving weak target responses; and 3) SSC introduces an auxiliary decoder optimized with contrastive learning to encourage the encoder to capture spectral deviation relationships, thereby improving adaptability to unseen scenes.

Accordingly, Section III-B introduces the proposed HADSim strategy, Section III-C presents the Sim2Real network architecture, and Section III-D describes the proposed 3C constraints in detail.

B. Data Simulation

Existing AD methods based on data augmentation or simulation [33], [34] rely on extracting background information from real-world data, which can easily lead to overfitting under data-scarce conditions. Although image patching strategies [35] alleviate this issue to some extent, they disrupt the global contextual structure of the image and require manual adjustment of patch sizes according to anomaly scales, thereby limiting their practicality and generalization capability. Moreover, as shown in Fig. 3, existing HAD methods are generally affected by variations in background spectral magnitude and local textures, as well as changes in the spatial morphology and spectral characteristics of anomalous targets and imaging noise, often leading to missed detections or false alarms.

To address these issues, this article proposes a HADSim strategy. The purpose of this simulation is not to strictly match the appearance of real remote sensing data, but rather to model key factors that are beneficial for improving HAD performance, including diverse backgrounds distributions, anomaly variations, and noise disturbances. The proposed strategy performs a simulation on the entire image (instead of local patches), enabling the modeling of rich background variations, perturbations in anomaly distributions, and noise effects. As a result, the model can exploit both spectral differences between anomalies and background as well as abundant contextual information, thereby enhancing its ability to discriminate complex background characteristics and improving robustness in real AD. Specifically, a HADSim method is proposed, which mainly consists of three key steps: background modeling, anomaly modeling, and noise simulation.

1) Background Modeling: First, a hyperspectral image $\mathbf{X}$ is initialized using spectra randomly sampled from the USGS spectral library.[1] Multiscale spatial distributions and spectral perturbations are then introduced to enhance background diversity. Finally, affine transformations and normalization are applied to improve robustness. Specifically, given a random parameter c (empirically $1 \le c \le 6$), c spectra are sampled from the USGS to form a background set $\mathbf{S}$. For each spectrum $\boldsymbol{P} \in \mathbf{S}$, spatial diversity and spectral perturbations are generated as follows.

Procedure 1: Spatial Diversity Generation. First, large blocks with random heights and widths (e.g., 64–128 pixels) are generated and filled with background spectra to simulate large-area land-cover distributions. Then, smaller blocks or stripes (e.g., 3–26 pixels) are introduced to model local edges and textures

$$\mathbf{X}(i, j, :) = \boldsymbol{P}, \quad (i, j) \in \mathbf{B}_b \tag{1}$$

where $\mathbf{B}_b$ denotes the generated background regions.

Procedure 2: Spectral Perturbation Generation. A random spectral reflectance scaling factor I_b is applied to block-shaped regions with random sizes (e.g., 32–64 pixels) to simulate local surface variations. Smaller stripe or block regions (e.g., 3×16 pixels) are then generated and similarly scaling-adjusted to mimic textured surface undulations

$$\mathbf{X}(i, j, :) = I_b \cdot \mathbf{X}(i, j, :), \quad (i, j) \in \mathbf{B}_p \tag{2}$$

where $\mathbf{B}_p$ represents the generated perturbation regions.

Procedure 3: Spatial Affine Transformation. Finally, the generated image $\mathbf{X}$ and its background label $\mathbf{L}_b$ undergo affine transformations–random rotation, deformation, translation, and scaling—to enhance spatial diversity while preserving spectral continuity

$$\mathbf{X}, \mathbf{L}_b = \mathcal{F}_{Affine}(\mathbf{X}, \mathbf{L}_b) \tag{3}$$

where $\mathcal{F}_{Affine}(\cdot)$ denotes the affine transformation operation.

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These steps simulate diverse background variations and generate rich spectral–spatial perturbations, helping improve the model's robustness to interference and AD accuracy.

2) Anomaly Modeling: In the anomaly modeling stage, anomaly targets are generated via anomaly injection and spectral modulation, resulting in diverse spatial–spectral patterns and deviations in spectral magnitude from the background. Specifically, anomaly spectra are generated by band permutation of randomly sampled spectra, with random-sized small blocks and spectral reflectance magnitude adjustments used to model spatial–spectral diversity.

Procedure 1: Spectral Generation. Spectra are randomly selected from the USGS, and anomaly spectra $\tilde{\boldsymbol{P}}$ are constructed through random band shuffling and spectral noise perturbations to simulate the spectral diversity of anomalies.

Procedure 2: Spatial Generation. Random block-shaped regions with random heights and widths (e.g., 1–8 pixels) are generated, into which the anomaly spectra are injected. Subsequently, affine transformations are applied to simulate different morphological variations of anomaly targets

$$\mathbf{X}, \mathbf{L}_a = \mathcal{F}_{Affine}\left(\mathbf{X}, \mathbf{L}_a\right) | \mathbf{X}\left(i, j, :\right) = \tilde{\boldsymbol{P}}, (i, j) \in \mathbf{B}_a \tag{4}$$

where $\mathbf{B}_a$ denotes the generated anomaly regions and $\mathbf{L}_a$ represents the anomaly labels.

Procedure 3: Anomaly Deviation Generation. A spectral reflectance scaling factor I_a is randomly selected to adjust the reflectance magnitude of the anomaly targets. The enhanced image is then normalized to generate tensor samples that meet the model input requirements. This process can be expressed as

$$\mathbf{X}\left(i, j, :\right) = I_a \cdot \mathbf{X}\left(i, j, :\right), (i, j) \in \mathbf{B}_a. \tag{5}$$

This strategy perturbs spectral and spatial structures to generate rich and controllable hyperspectral anomalies. Combined with diversified background samples, it yields representative, complex training data that enhance model generalization.

3) Noise Modeling: To simulate diverse noise in real hyperspectral scenes, random noise perturbations are applied during data modeling, enhancing the model's robustness and generalization. This module randomly selects noise types and parameters to generate statistically varied noise under different intensities. Four common types are considered: Gaussian, uniform, salt-and-pepper, and Poisson. Let the ideal hyperspectral image be $\mathbf{X} = [x_{ijl}]$, where x_{ijl} denotes the pixel value at spatial location (i, j) in the lth spectral band. The distributions of different noise types are described as follows.

Gaussian Noise: Gaussian noise is commonly used to model electronic thermal noise during sensor sampling, and its probability density function is given by

$$\mathcal{N}_{Gaussian}\left(x_{ijl}; \mu, \sigma^2\right) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{\left(x_{ijl} - \mu\right)^2}{2\sigma^2}\right) \tag{6}$$

where μ denotes the noise mean and σ is the standard deviation.

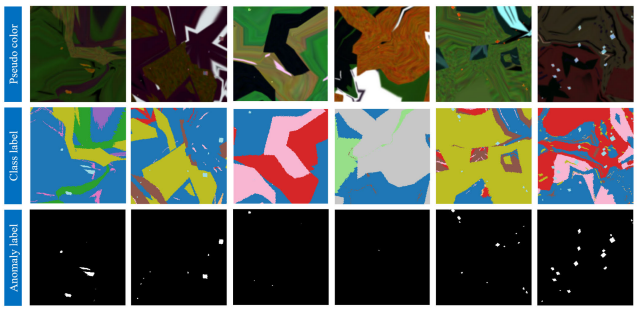


Fig. 4. Examples of data generated by HADSim. The first to third rows display the pseudo-color imagery, land-cover classification labels, and anomaly labels, respectively.

Uniform Noise: Uniform noise is often used to simulate nonuniform illumination or sensor response variations

$$\mathcal{N}_{Uniform}\left(x_{ijl}; a, b\right) = \begin{cases} \dfrac{1}{b-a}, & a \le n_{ijl} \le b \\ 0, & \text{otherwise} \end{cases} \tag{7}$$

where a and b control the amplitude range of the noise.

Salt-and-Pepper Noise: Salt-and-pepper noise is commonly used to simulate discrete extreme perturbations caused by transmission errors or pixel saturation

$$\mathcal{N}_{Salt\&Pepper}\left(x_{ijl}; p\right) = \begin{cases} 0, & \text{with probability } p/2 \\ 1, & \text{with probability } p/2 \\ x_{ijl}, & \text{with probability } 1-p \end{cases} \tag{8}$$

where p denotes the noise density, controlling the proportion of corrupted pixels.

Poisson Noise: Poisson noise is typically used to model photon-counting statistical fluctuations

$$\mathcal{N}_{Poisson}\left(n_{ijl} = k\right) = \frac{\lambda^k e^{-\lambda}}{k!} \tag{9}$$

where n_{ijl} and k denote the noisy and realized photon counts, respectively, and $\lambda = \text{scale} \times x_{ijl}$ represents the expected photon count proportional to the pixel reflectance.

Accordingly, the multinoise generation process adopted in this work can be uniformly expressed as

$$\begin{aligned} &\tilde{\mathbf{X}} = \mathbf{X} + \mathcal{N}_t\left(\mathbf{X}; \theta_t\right) \\ &t \in \left\{\mathcal{N}_{Gaussian}, \mathcal{N}_{Uniform}, \mathcal{N}_{Salt\&Pepper}, \mathcal{N}_{Poisson}\right\} \end{aligned} \tag{10}$$

where $\tilde{\mathbf{X}}$ denotes the noisy HSI, and t and θ_t denote the noise type and its parameters, respectively. In the noise generation process, one noise type and its corresponding parameter set (controlling noise strength) are randomly selected to generate $\tilde{\mathbf{X}}$, thereby improving model robustness. After noise injection, Gaussian smoothing suppresses isolated extremes and ensures spatial continuity. This random noise modeling enhances robustness and stability in real-world AD.

As shown in Fig. 4, the simulated data exhibit rich background structures, texture variations, and diverse anomaly distributions, effectively boosting the model's discrimination ability, as further validated in Section IV-C.

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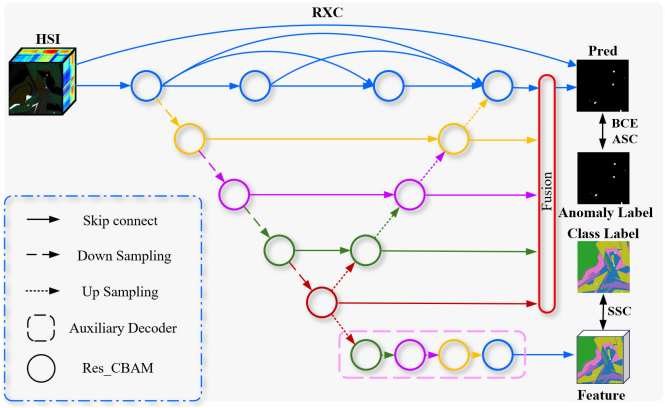


Fig. 5. Structure of the Sim2Real.

C. Sim2Real Network

Based on the simulated dataset, this article proposes an end-to-end model Sim2Real. As illustrated in Fig. 5, Sim2Real performs AD learning under the guidance of RX prior information. The model adopts U-Net as the backbone network to enable multiscale spatial context feature extraction. To mitigate the loss of fine-grained details in deep networks, dense connection structures are introduced at the original image resolution to enhance the preservation and propagation of pixel-level features. In addition, to realize SSC, an auxiliary decoder that does not participate in inference is introduced. During training, it serves as a spectral similarity regularization constraint through contrastive learning with class labels, forcing the encoder to learn generic spectral deviation relationships between anomalies and background rather than dataset-specific features. Specifically, Sim2Real uses the RX module to compute a prior, extracts and fuses multiscale features with a densely connected U-Net to generate an anomaly response map, and obtains the final output via residual fusion with the RX prior. During training, an auxiliary encoder branch with an SSC contrastive learning scheme is introduced for feature regularization. The framework consists of an encoder, a main decoder, an auxiliary decoder, a dense connection structure, and the RX module.

1) Encoder: The encoder extracts multiscale spatial features from the input hyperspectral image through five cascaded feature extraction stages with downsampling, enabling joint spatial–spectral modeling. Each stage adopts a standard residual block with two 3×3 convolutions, normalization, and ReLU activation (Res_CBAM [55]). Let the input hyperspectral image be $\mathbf{X}$. The encoder outputs $\mathbf{E}_0, \mathbf{E}_1, \mathbf{E}_2, \mathbf{E}_3, \mathbf{E}_4$, where $\mathbf{E}_0 = \mathcal{F}_{Res}(\mathbf{X})$, and $\mathbf{E}_4$ is the lowest-resolution semantic feature. The encoding process is

$$\mathbf{E}_i = \mathcal{P}_{Max}\left(\mathcal{F}_{Res}\left(\mathbf{E}_{i-1}\right)\right), i = 1, 2, 3, 4 \tag{11}$$

where $\mathcal{F}_{Res}(\cdot)$ denotes Res_CBAM convolutional feature extraction and $\mathcal{P}_{Max}(\cdot)$ is downsampling with stride 2.

2) Dense Connections: To prevent small-target feature degradation caused by deep convolutions, a dense connection module is introduced at the original-resolution feature stage to reuse shallow details. The module contains three residual blocks with outputs $\mathbf{C}_1, \mathbf{C}_2, \mathbf{C}_3$. For the jth layer

$$\mathbf{C}_j = \mathcal{F}_{Res}\left(\left[\mathbf{C}_0, \ldots, \mathbf{C}_{j-1}\right]\right), j = 1, 2, 3 \tag{12}$$

where $\mathbf{C}_0 = \mathbf{E}_0$, and $[\cdot]$ denotes channel-wise concatenation.

3) Main Decoder and Multiscale Fusion: The main decoder reconstructs the anomaly response map via progressive upsampling and cross-scale fusion using skip connections. Let decoder features be $\mathbf{D}_0, \mathbf{D}_1, \ldots, \mathbf{D}_4$ with $\mathbf{D}_0 = \mathbf{E}_4$. The ith decoding stage is

$$\mathbf{D}_i = \mathcal{F}_{Res}\left(\left[\mathbf{E}_{5-i}, \mathcal{U}\left(\mathbf{D}_{i-1}\right)\right]\right), i = 1, 2, 3, 4 \tag{13}$$

where $\mathcal{U}(\cdot)$ denotes upsampling. The anomaly response map is generated through multiscale fusion

$$\mathbf{A}_{Net} = \mathcal{F}_{Res}\left(\left[\mathcal{U}\left(\mathbf{D}_0\right), \mathcal{U}\left(\mathbf{D}_1\right), \ldots, \mathbf{D}_4\right]\right). \tag{14}$$

4) RX Statistical Prior and Residual Fusion: To introduce statistical anomaly constraints, the RX module produces a prior response $\mathbf{A}_{RX}$. The final prediction is obtained by residual fusion

$$\hat{\mathbf{A}} = \mathbf{A}_{Net} + \mathbf{A}_{RX}. \tag{15}$$

This residual guidance stabilizes detection and improves cross-scene generalization.

5) Auxiliary Decoder: The auxiliary decoder shares encoder features but has an independent objective to regularize representations via SSC contrastive learning. Let auxiliary features be $\mathbf{d}_0, \mathbf{d}_1, \ldots, \mathbf{d}_4$ with $\mathbf{d}_0 = \mathbf{E}_4$. The recursion is

$$\mathbf{d}_i = \mathcal{F}_{Res}\left(\mathcal{U}\left(\mathbf{d}_{i-1}\right)\right), i = 1, 2, 3, 4. \tag{16}$$

After training, the learned features impose spectral-similarity consistency on the encoder representations.

D. 3C

1) RXC: End-to-end HAD networks often lack explicit global statistical modeling, which makes them prone to misinterpreting local saliency or spectral reflectance variations as anomalies. As a result, such models are highly sensitive to high-reflectance background regions or spurious targets, often producing large-scale false alarms, especially when anomaly samples are scarce and training constraints are insufficient. RX-based AD measures the degree of anomaly by characterizing the distance between pixel spectra and a global statistical model. Its statistical properties are effective in suppressing background-induced false alarms; however, its responses to true anomalies are often not sufficiently discriminative. Consequently, most existing methods adopt a two-stage strategy of background reconstruction followed by RX detection [33], [34]. Nevertheless, this decoupled pipeline lacks collaborative constraints between statistical priors and feature learning, making it difficult to learn feature representations that are truly beneficial for RX-based discrimination.

To address these issues, this article explicitly incorporates RX statistical priors into the HAD network and performs residual learning on an RX preliminary anomaly response. This design guides the network to further enhance the discrimination between anomalies and background. As illustrated in

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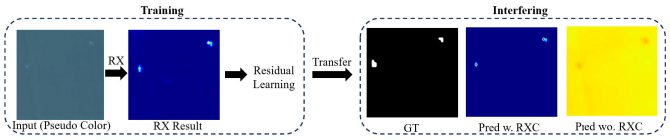


Fig. 6. Design motivation of RXC and a comparison of detection results with (w.) and without (wo.) RXC.

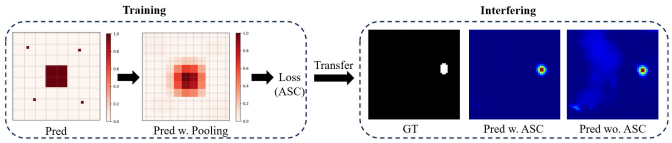


Fig. 7. Design motivation of ASC and a comparison of detection results with (w.) and without (wo.) ASC.

Fig. 5, the RX computation first produces an initial anomaly map, which is then integrated into the network via residual connections, thereby effectively improving AD performance under the guidance of statistical priors.

As shown in Fig. 6, under low anomaly saliency, removing the RXC constraint causes high background responses, while RX suppresses them. RX-based residual learning further improves anomaly–background contrast, demonstrating the effectiveness of RXC. Specifically, let the input hyperspectral image be denoted as $\mathbf{X} \in \mathbb{R}^{H\times W\times L}$, where L represents the number of spectral bands and $H \times W$ denotes the spatial resolution. The image is reshaped into a set of pixel vectors $\mathbf{x} = [\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_N]$, where $N = H \times W$ and $\boldsymbol{x}_n \in \mathbb{R}^L$ denotes the spectral vector of the nth pixel. First, the global spectral mean vector is computed as

$$\boldsymbol{\mu} = \frac{1}{N}\sum_{n=1}^{N} \boldsymbol{x}_n. \tag{17}$$

Then, the centralized spectral matrix is constructed as

$$\mathbf{Z} = \left[\boldsymbol{x}_1 - \boldsymbol{\mu}, \ldots, \boldsymbol{x}_N - \boldsymbol{\mu}\right]. \tag{18}$$

Accordingly, the covariance matrix is defined as

$$\boldsymbol{\Sigma} = \frac{1}{N-1}\mathbf{Z}\mathbf{Z}^{\top} + \varepsilon\mathbf{I} \tag{19}$$

where ε is a numerical stabilization term and $\mathbf{I}$ denotes the identity matrix.

Based on the above formulation, the RX anomaly measure for the nth pixel is given by

$$r_n = (\boldsymbol{x}_n - \boldsymbol{\mu})^{\top} \boldsymbol{\Sigma}^{-1} (\boldsymbol{x}_n - \boldsymbol{\mu}) \tag{20}$$

where r_n denotes the RX anomaly response value of the nth pixel. The RX responses of all pixels are then rearranged into a spatial anomaly map $\mathbf{A}_{RX} \in \mathbb{R}^{H\times W}$.

2) ASC: Considering that existing AD methods [56], [57], [58] based on low-rank sparse modeling predominantly rely on pixel-level sparsity regularization (e.g., $\mathcal{L}_1/\mathcal{L}_2$), their constraints mainly act on the response magnitude of individual pixels. As a result, they struggle to distinguish large-area false anomaly responses from genuine small-target anomalies at the level of spatial structure, which often leads to spatial diffusion of anomaly responses or excessive suppression of small targets.

To address this issue, we introduce a block-level response suppression constraint based on spatial average downsampling into the AD model. Specifically, the anomaly response map is first subjected to block-wise average downsampling, aggregating pixel-level responses into local average intensities and elevating the spatial constraint from the pixel level to the block level. Subsequently, a nonlinear regularization constraint is imposed on the downsampled block-level response map to suppress the spatial expansion of anomaly responses.

As illustrated in Fig. 7, large-area anomaly responses with high spatial occupancy tend to preserve relatively high responses across multiple blocks after downsampling and are therefore more strongly suppressed during training. Crucially, the downsampling operation not only reduces resolution but also aggregates local responses, enabling the loss to become sensitive to the spatial occupancy of anomaly responses. In contrast, sparse small-target anomalies are more easily diluted during block-wise averaging and thus produce relatively weak block-level responses after downsampling, resulting in limited penalties after quadratic regularization. Unlike conventional pixel-level sparsity regularization, the proposed constraint achieves structured regularization that is sensitive to the spatial occupancy of anomaly responses through block-wise downsampling and nonlinear response penalization. Consequently, the penalty strength becomes correlated with the spatial occupancy and expansion degree of anomaly responses, enabling effective suppression of large-area false anomalies while better preserving small-target anomalies. Specifically, let the anomaly response map output by the model be denoted as $\hat{\mathbf{A}} \in \mathbb{R}^{H\times W}$. It is mapped to a low-resolution space via an average downsampling operator $\mathcal{P}^k_{Avg}(\cdot)$

$$\tilde{\mathbf{A}} = \mathcal{P}^k_{Avg}\left(\hat{\mathbf{A}}\right) \tag{21}$$

where $\mathcal{P}^k_{Avg}(\cdot)$ denotes an average pooling operation with kernel size $k \times k$ and stride k. Subsequently, a quadratic block-response suppression constraint is defined in the downsampled space as

$$\mathcal{L}_{ASC} = \frac{1}{N}\sum_{i=1}^{H/k}\sum_{j=1}^{W/k} \tilde{\mathbf{A}}^2_{i,j} \tag{22}$$

where $N = (H/k)(W/k)$ is the number of downsampling units. This constraint suppresses the quadratic energy of block-level responses. Large-area anomalies tend to preserve relatively high responses across multiple blocks after downsampling and therefore receive stronger penalties after quadratic regularization. In contrast, sparse small targets are more easily diluted during spatial aggregation and thus incur relatively weaker penalties. Consequently, $\mathcal{L}_{ASC}$ suppresses large false alarms while better preserving small-target anomalies. The overall loss function is defined as

$$\mathcal{L} = \mathcal{L}_{Task} + \alpha\mathcal{L}_{ASC} \tag{23}$$

where $\mathcal{L}_{Task}$ denotes the primary task loss and α is a weighting coefficient.

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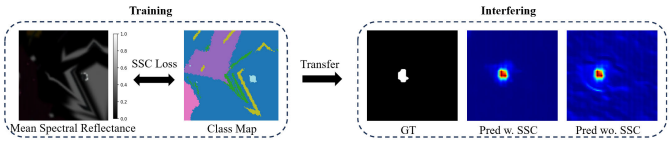


Fig. 8. Design motivation of SSC and a comparison of detection results with (w.) and without (wo.) SSC.

3) SSC: Most existing AD methods [53], [59], [60], [61] primarily focus on learning specific background representations from training data, rather than exploring the general deviation relationships between anomalies and backgrounds. This paradigm makes models prone to overfitting the background distribution of the training set and limits their ability to identify anomalies in unseen scenes through an understanding of spectral differences among pixels, thereby restricting generalization performance.

To address this issue, we introduce a spectral similarity regularization mechanism. By incorporating an auxiliary decoder together with a contrastive learning strategy, this mechanism guides the encoder to learn feature representations based on interpixel spectral relationships, thereby shifting the detection logic from "matching known backgrounds" to "identifying spectral deviations," and enhancing the model's adaptability to novel scenes. As illustrated in Fig. 8, for variations in spectral reflectance magnitude (e.g., local surface undulations) within the same background region, the proposed constraint enforces high consistency among these pixels in the decoder feature space, while maintaining clear separation from pixels in heterogeneous regions. In this way, the network is driven to learn the structural similarity of spectra, thereby helping suppress false alarms caused by noise or complex background variations.

Specifically, a contrastive loss $\mathcal{L}_{SSC}$ is introduced using the auxiliary decoder features and their class labels, encouraging the encoder to distinguish spectral discrepancies between anomalies and background while reducing reliance on specific background characteristics. This auxiliary pathway guides the encoder to focus on interpixel spectral deviations without altering the main detection logic, leading to more spectrally general features. Acting as both constraint and guidance during training, it improves stability and generalization under unknown backgrounds. The loss $\mathcal{L}_{SSC}$ measures the spectral structural similarity of the auxiliary decoder features, and the overall regularization loss is formulated as

$$\mathcal{L}_{SSC} = \frac{1}{N^2}\sum_{i=1}^{N-1}\sum_{j=i+1}^{N}\left(\hat{S_{ij}} - S_{ij}\right)^2 = \frac{1}{N^2}\|\hat{S} - S\|_F^2 \tag{24}$$

where N denotes the number of sampled pixels. Pixel sampling is used to reduce computational cost, and N is empirically set to 2048 in this work. Here, $\hat{S}_{ij}$ denotes the feature correlation, while S_{ij} denotes the binary similarity label. The feature correlation $\hat{S}_{ij}$ is computed as

$$\hat{S}_{ij} = \frac{\boldsymbol{F}_i^\top \boldsymbol{F}_j}{\|\boldsymbol{F}_i^\top\|_2 \|\boldsymbol{F}_j\|_2} \tag{25}$$

where $\boldsymbol{F}$ represents pixel-wise features sampled from the auxiliary decoder output $\mathbf{d}_4$. The binary similarity S_{ij} is defined as

$$S_{ij} = \begin{cases} 1, & L_i = L_j \\ 0, & \text{otherwise} \end{cases} \tag{26}$$

where L denotes the land-cover class label corresponding to feature $\boldsymbol{F}$ at the spatial location.

By minimizing $\mathcal{L}_{SSC}$, the model enforces spectral consistency among different materials in the feature space, improving anomaly segmentation accuracy and reducing background false alarms.

Finally, the overall loss function is defined as

$$\mathcal{L} = \mathcal{L}_{Task} + \beta\mathcal{L}_{SSC} \tag{27}$$

where $\mathcal{L}_{Task}$ denotes the AD task loss of the main decoder, and β is a weighting factor that balances the contribution of the spectral similarity regularization.

E. Loss Function

To achieve high-precision AD, three loss functions are jointly employed during training: binary cross-entropy anomaly loss, the proposed mean downsampling sparsity loss, and contrastive loss. These losses collaboratively constrain the model's discriminative capability, spatial sparsity, and feature separability. The overall loss function is defined as

$$\mathcal{L}_{Total} = \mathcal{L}_{Task} + \alpha\mathcal{L}_{ASC} + \beta\mathcal{L}_{SSC} \tag{28}$$

where $\mathcal{L}_{Task} = \mathcal{L}_{BCE}$ denotes the binary cross-entropy AD loss. The coefficients α and β are balancing factors for the corresponding loss terms.

The binary cross-entropy anomaly loss is used to supervise pixel-level anomaly classification by measuring the cross-entropy between the predicted probability distribution and the ground-truth label distribution [35], [62]. It enforces accurate discrimination between anomalous and background pixels. The loss is defined as

$$\mathcal{L}_{BCE} = -\frac{1}{N}\sum_{i=1}^{N}\left[\boldsymbol{y}_i \log\left(\boldsymbol{p}_i\right) + \left(1 - \boldsymbol{y}_i\right)\log\left(1 - \boldsymbol{p}_i\right)\right] \tag{29}$$

where $\boldsymbol{p}_i$ denotes the predicted probability that the ith pixel is anomalous, $\boldsymbol{y}_i$ is the corresponding ground-truth label, and N represents the total number of pixels. This loss ensures the pixel-level anomaly discrimination capability of the model.

IV. Experiments

In this section, we first introduce the experimental datasets, evaluation metrics, and implementation details. Then, the proposed Sim2Real-HAD is compared with several representative HAD methods. Finally, comprehensive model analyses are conducted to verify the effectiveness of each component and parameter setting in Sim2Real-HAD.

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TABLE I
SCENES AND INFORMATION OF THE ABU DATASET

Scene	Image ID	Location	Resolution	Sensor
Airport	a	Los Angeles	7.1m	AVIRIS
	b	Los Angeles	7.1m	AVIRIS
	c	Los Angeles	7.1m	AVIRIS
	d	Gulfport	3.4m	AVIRIS
Beach	a	Champagne Bay	4.4m	AVIRIS
	b	Pavia	1.3m	ROSIS-03
	c	San Diego	7.5m	AVIRIS
	d	Cat Island	17.2m	AVIRIS
Urban	a	Texas Coast	17.2m	AVIRIS
	b	Texas Coast	17.2m	AVIRIS
	c	Gainesville	3.5m	AVIRIS
	d	Los Angeles	7.1m	AVIRIS
	e	Los Angeles	7.1m	AVIRIS

A. Experimental Setup

1) Datasets: The test datasets consist of 113 publicly available hyperspectral scenes, including 100 scenes from HAD100 and 13 scenes from ABU (airport, beach, and urban). Specifically, all images in the ABU dataset have a spatial resolution of 100×100 pixels. Table I summarizes the acquisition locations, spatial resolutions, and sensor information of these three datasets.

HAD100 is a recently released real-world dataset by Li et al. [33], acquired using the Airborne Visible/Infrared Imaging Spectrometer-Next Generation (AVIRIS-NG). All images have a spatial resolution of 64×64 pixels with 276 effective spectral bands, and the background types cover diverse land-cover scenes, including forests, deserts, farmlands, and rivers. In addition, to further validate the effectiveness of the proposed HADSim background simulation strategy, the AVIRIS-NG background training set provided by Li et al. [33] was additionally used for background replacement evaluation. This dataset contains 260 background images.

2) Compared Methods and Evaluation Metrics: We compare the proposed method with eight representative HAD approaches, including traditional methods RX [7], FrFT [63], and FEBPAD [53]; scene-dependent background reconstruction-based methods Auto-AD [39] and GIL [54]; and one-step training methods AETNet [33], FEN [34], and TDD [35].

For quantitative comparison, five widely used AD metrics are adopted [35], [41], [64], [65]. Let $\mathrm{AUC}_{(\mathrm{D,F})}$ denote the standard ROC-AUC, $\mathrm{AUC}_{(\mathrm{D},\tau)}$ the integrated detection energy, and $\mathrm{AUC}_{(\mathrm{F},\tau)}$ the integrated false alarm energy over thresholds. The adopted metrics include $\mathrm{AUC}_{(\mathrm{D,F})} \uparrow$ (detection performance), $\mathrm{AUC}_{(\mathrm{F},\tau)} \downarrow$ (false alarm energy), $\mathrm{AUC}_{\mathrm{BS}} \uparrow$ (background suppression capability $[\mathrm{AUC}_{(\mathrm{D,F})} - \mathrm{AUC}_{(\mathrm{F},\tau)}]$), $\mathrm{AUC}_{\mathrm{SNPR}} \uparrow$ (signal-to-noise discrimination ratio $[\mathrm{AUC}_{(\mathrm{D},\tau)}/\mathrm{AUC}_{(\mathrm{F},\tau)}]$), and $\mathrm{AUC}_{\mathrm{OADP}} \uparrow$ (comprehensive metric $[\mathrm{AUC}_{(\mathrm{D,F})} + \mathrm{AUC}_{(\mathrm{D},\tau)} + 1 - \mathrm{AUC}_{(\mathrm{F},\tau)}]$).

3) Implementation Details: All methods are implemented in Python and trained and evaluated on a workstation equipped with an 11th Gen Intel[2] Core[3] i7-11700K at 3.60 GHz CPU and an NVIDIA RTX 3060 Ti GPU (8 GB memory).

For the proposed Sim2Real-HAD, the simulated training images, test images, and network input size are uniformly set or resized to 256×256, with the number of spectral bands fixed at 64. Training uses the Adam optimizer with an initial learning rate of 5×10^{-2}, which is decayed by a factor of 10 after 100 epochs. The batch size is 4, and the total number of training epochs is 500. For a fair comparison, the spatial resolution of all methods is unified, while the number of spectral bands follows each method's original configuration. All competing methods are implemented and tuned according to their original papers, and, when available, evaluated using the authors' released pretrained weights to ensure optimal performance.

B. Model Comparison

1) Quantitative and Qualitative Comparison of Results: Tables II and III present the comparative results of different methods on the HAD100 and ABU datasets, respectively. It can be observed that Sim2Real achieves two best and three second-best results among the five key metrics on HAD100, demonstrating its superior HAD performance. Furthermore, on the ABU dataset, Sim2Real achieves three best results and one third-best result among all competing methods. Overall, it also attains the best overall performance on the ABU dataset, further demonstrating the effectiveness of the proposed method. More importantly, across a total of 113 scenes, competing methods obtain optimal results only in specific local scenarios, whereas Sim2Real consistently performs well across different scenes, exhibiting stronger generalization capability.

Fig. 9 illustrates the HAD response maps produced by different methods in representative detection scenarios, where clear differences can be observed. Although TDD is highly sensitive to anomalous regions, it also produces a large number of false alarms and exhibits excessive responses to high-reflectance spectral components. Other methods, such as RX and FrFT, also generate considerable false positives in many scenarios. Overall, the proposed Sim2Real method achieves the best tradeoff across diverse scenes, exhibiting both strong background suppression capability and pronounced anomaly responses.

Finally, Fig. 10 shows the ROC curves of different methods in some detection scenarios. It can be seen that the proposed Sim2Real consistently demonstrates stable and superior performance across different scenes. Compared with RX, FrFT, and various deep learning-based methods, the ROC (PD-FA) curves of Sim2Real are consistently closer to the upper-left corner, indicating higher detection probabilities at lower false-alarm rates and thus stronger anomaly–background discrimination capability. Moreover, the threshold-PD and threshold-FA curves reveal that Sim2Real maintains a high detection rate over a wide range of thresholds while rapidly

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TABLE II

AUC RESULTS OF DIFFERENT AD METHODS ON HAD100, ↑ INDICATES HIGHER IS BETTER, WHILE ↓ INDICATES LOWER IS BETTER. BOLD **BLACK** (THE BEST), RED (THE SECOND BEST), AND BLUE (THE THIRD BEST). HAD100 IS SPLIT INTO GROUP1–GROUP10; COLUMNS WITH METHOD NAMES AS HEADERS SHOW GROUP-WISE AVERAGES, WHILE THE AVG ROW DENOTES THE OVERALL AVERAGE

Images	Metrics	RX	FrFT	FEBP-AD	Auto-AD	GIL	AETNet	FEN	TDD	Sim2Real
Avg	$AUC_{(D,F)}\uparrow$	0.9654	0.9831	0.9620	0.9545	0.9426	**0.9906**	0.9890	0.7571	0.9893
	$AUC_{(F,\tau)}\downarrow$	0.0786	0.0396	0.0181	0.0139	0.0148	0.0588	0.0373	0.4206	**0.0048**
	$AUC_{BS}\uparrow$	0.8868	0.9435	0.9439	0.9406	0.9278	0.9317	0.9518	0.3365	**0.9846**
	$AUC_{SNPR}\uparrow$	6.1203	11.087	**1139.0**	52.831	56.808	10.364	10.872	5.3095	131.12
	$AUC_{ODP}\uparrow$	2.2576	2.2791	2.1366	2.2184	2.2007	**2.3215**	2.1958	2.0270	2.3184
Group1	$AUC_{(D,F)}\uparrow$	0.9944	0.9954	0.9752	0.9879	0.9907	**0.9991**	0.9978	0.7688	0.9968
	$AUC_{(F,\tau)}\downarrow$	0.0741	0.0352	0.0085	0.0113	0.0103	0.0639	0.0393	0.3992	**0.0046**
	$AUC_{BS}\uparrow$	0.9202	0.9602	0.9667	0.9766	0.9804	0.9352	0.9585	0.3696	**0.9923**
	$AUC_{SNPR}\uparrow$	6.2956	12.257	**298.48**	32.559	36.234	9.7833	11.245	4.4394	120.16
	$AUC_{ODP}\uparrow$	2.3377	2.3450	2.1841	2.3087	2.3133	**2.4104**	2.2434	2.0377	2.3967
Group2	$AUC_{(D,F)}\uparrow$	0.9633	0.9968	0.9747	0.9734	0.9620	**0.9986**	0.9959	0.8805	0.9969
	$AUC_{(F,\tau)}\downarrow$	0.0883	0.0363	0.0234	0.0105	0.0117	0.0727	0.0446	0.3480	**0.0046**
	$AUC_{BS}\uparrow$	0.8750	0.9605	0.9513	0.9629	0.9503	0.9259	0.9513	0.5325	**0.9924**
	$AUC_{SNPR}\uparrow$	5.3240	12.835	**4959.3**	70.902	58.483	7.8677	8.2434	3.7036	107.18
	$AUC_{ODP}\uparrow$	2.2813	2.3705	2.1815	2.3502	2.3240	**2.3848**	2.2481	2.2568	2.3787
Group3	$AUC_{(D,F)}\uparrow$	0.9918	**0.9989**	0.9655	0.9591	0.9687	0.9979	0.9968	0.7387	0.9961
	$AUC_{(F,\tau)}\downarrow$	0.0741	0.0401	0.0228	0.0103	0.0120	0.0503	0.0233	0.3792	**0.0043**
	$AUC_{BS}\uparrow$	0.9177	0.9588	0.9427	0.9488	0.9567	0.9476	0.9735	0.3596	**0.9919**
	$AUC_{SNPR}\uparrow$	9.2482	14.383	108.17	94.494	123.90	14.539	14.879	2.7142	**155.13**
	$AUC_{ODP}\uparrow$	2.3613	2.4337	2.1491	2.2726	2.3004	**2.4353**	2.2234	2.0435	2.4122
Group4	$AUC_{(D,F)}\uparrow$	0.9656	0.9889	0.9818	0.9058	0.9217	**0.9954**	0.9923	0.7522	0.9940
	$AUC_{(F,\tau)}\downarrow$	0.0623	0.0357	0.0138	0.0145	0.0146	0.0488	0.0350	0.2794	**0.0044**
	$AUC_{BS}\uparrow$	0.9033	0.9531	0.9680	0.8913	0.9070	0.9466	0.9573	0.4727	**0.9896**
	$AUC_{SNPR}\uparrow$	4.7388	9.4672	**295.97**	23.746	26.164	16.225	9.1589	4.4919	88.928
	$AUC_{ODP}\uparrow$	2.1879	2.2525	2.2564	2.0749	2.1166	2.2607	2.1655	2.0383	**2.2704**
Group5	$AUC_{(D,F)}\uparrow$	0.9405	**0.9495**	0.9141	0.9389	0.9255	0.9467	0.9493	0.4796	0.9388
	$AUC_{(F,\tau)}\downarrow$	0.1000	0.0517	0.0172	0.0254	0.0216	0.0842	0.0534	0.6393	**0.0083**
	$AUC_{BS}\uparrow$	0.8406	0.8978	0.8968	0.9136	0.9039	0.8625	0.8959	0.1597	**0.9306**
	$AUC_{SNPR}\uparrow$	4.5677	7.4899	**249.13**	20.573	23.085	6.6654	5.5225	1.1990	91.086
	$AUC_{ODP}\uparrow$	2.2415	2.2313	2.0687	2.1830	2.1663	**2.3014**	2.1386	1.5197	2.2727
Group6	$AUC_{(D,F)}\uparrow$	0.9617	0.9739	0.9659	0.9727	0.9746	**0.9970**	0.9947	0.6761	0.9966
	$AUC_{(F,\tau)}\downarrow$	0.0880	0.0577	0.0255	0.0227	0.0239	0.0665	0.0597	0.5439	**0.0055**
	$AUC_{BS}\uparrow$	0.8736	0.9163	0.9404	0.9500	0.9508	0.9304	0.9349	0.1322	**0.9911**
	$AUC_{SNPR}\uparrow$	5.0757	8.5916	**493.03**	21.321	24.135	8.4801	8.3894	2.0526	81.642
	$AUC_{ODP}\uparrow$	2.2327	2.2522	2.1484	2.2432	2.2401	**2.3031**	2.2406	1.8685	2.2991
Group7	$AUC_{(D,F)}\uparrow$	0.9433	0.9846	0.9577	0.9724	0.9626	0.9980	**0.9981**	0.9603	0.9962
	$AUC_{(F,\tau)}\downarrow$	0.0795	0.0408	0.0224	0.0060	0.0062	0.0353	0.0203	0.2744	**0.0033**
	$AUC_{BS}\uparrow$	0.8638	0.9438	0.9354	0.9664	0.9565	0.9628	0.9778	0.6860	**0.9929**
	$AUC_{SNPR}\uparrow$	6.0232	7.7104	28.034	82.885	72.889	15.369	20.933	24.766	**113.94**
	$AUC_{ODP}\uparrow$	2.2235	2.2288	2.0890	2.2363	2.2246	2.3361	2.1946	**2.3842**	2.2805
Group8	$AUC_{(D,F)}\uparrow$	0.9468	0.9737	0.9653	0.9609	0.9624	0.9902	**0.9921**	0.6552	0.9893
	$AUC_{(F,\tau)}\downarrow$	0.0703	0.0390	0.0137	0.0118	0.0119	0.0713	0.0503	0.5539	**0.0060**
	$AUC_{BS}\uparrow$	0.8766	0.9346	0.9517	0.9492	0.9504	0.9188	0.9418	0.1013	**0.9833**
	$AUC_{SNPR}\uparrow$	8.6358	16.558	67.416	60.052	50.944	8.7609	14.773	3.6993	**201.79**
	$AUC_{ODP}\uparrow$	2.2980	2.2520	2.1078	2.2522	2.2499	2.3373	2.2784	1.8728	**2.3788**
Group9	$AUC_{(D,F)}\uparrow$	0.9613	0.9810	0.9314	0.9499	0.8885	0.9927	**0.9934**	0.7600	0.9926
	$AUC_{(F,\tau)}\downarrow$	0.0956	0.0284	0.0177	0.0102	0.0164	0.0454	0.0317	0.3989	**0.0027**
	$AUC_{BS}\uparrow$	0.8657	0.9526	0.9137	0.9397	0.8722	0.9474	0.9615	0.3611	**0.9898**
	$AUC_{SNPR}\uparrow$	6.0494	13.186	**4872.3**	103.45	137.94	10.762	11.390	3.9345	303.02
	$AUC_{ODP}\uparrow$	2.2064	2.2205	2.0845	2.2115	2.1250	2.2612	2.1988	1.9633	**2.2901**
Group10	$AUC_{(D,F)}\uparrow$	0.9853	0.9882	0.9882	0.9239	0.8691	0.9901	0.9799	0.8991	**0.9959**
	$AUC_{(F,\tau)}\downarrow$	0.0533	0.0308	0.0161	0.0163	0.0196	0.0499	0.0146	0.3895	**0.0041**
	$AUC_{BS}\uparrow$	0.9320	0.9574	0.9722	0.9076	0.8494	0.9402	0.9653	0.5096	**0.9917**
	$AUC_{SNPR}\uparrow$	5.2448	8.3944	18.554	18.323	14.296	5.1910	4.1849	2.0940	**51.268**
	$AUC_{ODP}\uparrow$	2.2057	2.2047	2.0967	2.0518	1.9468	2.1843	2.0273	**2.2853**	2.2046

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TABLE III

AUC RESULTS OF DIFFERENT AD METHODS ON ABU, ↑ INDICATES HIGHER IS BETTER, WHILE ↓ INDICATES LOWER IS BETTER. BOLD **BLACK** (THE BEST), RED (THE SECOND BEST), AND BLUE (THE THIRD BEST). ABU IS SPLIT INTO THREE SCENE CLASSES; COLUMNS WITH METHOD NAMES AS HEADERS SHOW CLASS-WISE AVERAGES, WHILE THE AVG ROW DENOTES THE OVERALL AVERAGE

Images	Metrics	RX	FrFT	FEBP-AD	Auto-AD	GIL	AETNet	FEN	TDD	Sim2Real
Avg	$AUC_{(D,F)}\uparrow$	0.9455	0.9687	0.9370	0.9624	0.9609	**0.9817**	0.9786	0.7487	0.9766
	$AUC_{(F,\tau)}\downarrow$	0.0315	0.0244	0.0115	0.0074	0.0068	0.0214	0.0138	0.4336	**0.0038**
	$AUC_{BS}\uparrow$	0.9141	0.9443	0.9255	0.9550	0.9541	0.9603	0.9648	0.3152	**0.9727**
	$AUC_{SNPR}\uparrow$	5.7374	7.5014	39.991	73.461	80.127	9.3844	10.984	2.2811	**91.633**
	$AUC_{ODP}\uparrow$	2.0741	2.1024	2.0437	2.1870	**2.2293**	2.1275	2.0811	1.9673	2.1076
Airport	$AUC_{(D,F)}\uparrow$	0.8935	0.9518	0.8844	0.9255	0.9307	**0.9767**	0.9724	0.8578	0.9644
	$AUC_{(F,\tau)}\downarrow$	0.0331	0.0170	0.0043	0.0136	0.0120	0.0128	0.0127	0.3815	**0.0039**
	$AUC_{BS}\uparrow$	0.8604	0.9348	0.8801	0.9119	0.9187	**0.9639**	0.9597	0.4763	0.9606
	$AUC_{SNPR}\uparrow$	3.4394	5.8111	**33.088**	11.892	15.671	9.5464	10.789	2.6073	25.405
	$AUC_{ODP}\uparrow$	1.9638	2.0245	1.9619	2.0824	2.1187	2.0687	2.0615	**2.1242**	2.0373
Beach	$AUC_{(D,F)}\uparrow$	0.9678	0.9705	0.9619	**0.9837**	0.9824	0.9793	0.9738	0.6235	0.9765
	$AUC_{(F,\tau)}\downarrow$	0.0327	0.0347	0.0228	0.0044	0.0048	0.0330	0.0123	0.4160	**0.0037**
	$AUC_{BS}\uparrow$	0.9351	0.9358	0.9391	**0.9793**	0.9776	0.9463	0.9615	0.2075	0.9728
	$AUC_{SNPR}\uparrow$	8.4704	8.5746	6.8678	**106.40**	79.099	7.1219	10.281	2.0433	72.062
	$AUC_{ODP}\uparrow$	2.1864	2.2108	2.1008	2.2592	**2.3186**	2.2139	2.0605	1.7674	2.1699
Urban	$AUC_{(D,F)}\uparrow$	0.9753	0.9838	0.9648	0.9780	0.9697	0.9892	**0.9897**	0.7650	0.9889
	$AUC_{(F,\tau)}\downarrow$	0.0286	0.0214	0.0075	0.0041	**0.0037**	0.0185	0.0163	0.5033	0.0039
	$AUC_{BS}\uparrow$	0.9467	0.9624	0.9573	0.9739	0.9660	0.9707	0.9734	0.2617	**0.9848**
	$AUC_{SNPR}\uparrow$	5.3024	8.1186	80.017	102.09	145.61	11.485	11.881	2.1927	**177.43**
	$AUC_{ODP}\uparrow$	2.0722	2.0718	2.0683	2.2193	**2.2507**	2.1000	2.1213	2.0104	2.1155

TABLE IV

COMPARISON OF PARAMETERS, FLOPS, AND EXECUTION TIME OF DIFFERENT MODELS

Method	Params (M)	FLOPs (G)	Times (s)
RX	-	-	**0.01**
FrFT	-	-	63.9
FEBPAD	-	-	47.3
Auto-AD	3.03	-	16.0
GIL	3.55	-	46.2
AETNet	1.73	**15.7**	0.51
FEN	**0.68**	47.3	0.43
TDD	46.6	237	0.88
Sim2Real	0.77	45.2	0.29

suppressing false alarms, indicating low sensitivity to threshold selection and strong robustness. These observations remain consistent across most scenes, demonstrating that Sim2Real can stably extract discriminative anomaly responses under complex backgrounds and cross-scene conditions, thereby exhibiting excellent detection capability and generalization performance.

2) Model Efficiency Comparison: Table IV provides a detailed comparison of average runtime and model complexity on the HAD100 dataset. For traditional methods, no trainable parameters are involved, so Params and FLOPs are not reported. RX achieves fast detection speed, whereas FrFT and FEBPAD require significantly more time due to complex frequency domain or optimization computations. For deep learning methods, unsupervised models such as Auto-AD and GIL require iterative reconstruction-based parameter optimization for each scene, resulting in lower overall efficiency than one-shot inference models. Within the one-shot training models, Sim2Real achieves the fastest detection while maintaining a relatively low parameter scale (0.77M) and moderate computational load. Although its FLOPs are slightly higher than AETNet and its parameters slightly higher than FEN, its practical runtime efficiency is superior, demonstrating that Sim2Real strikes a more reasonable balance between model complexity and detection speed. This also highlights the efficiency advantage of end-to-end HAD models.

C. Model Analysis

In this section, model analysis experiments are conducted on the HAD100 dataset. All performance metrics are averaged over 100 scenes to evaluate the performance of different variants.

1) Effectiveness of the Simulation Strategy: Different simulation generation settings were tested to validate the effectiveness of the proposed HADSim strategy.

1) *Without BM (Background Modeling):* During data simulation, background disturbances such as large-scale and local texture fluctuations were removed.

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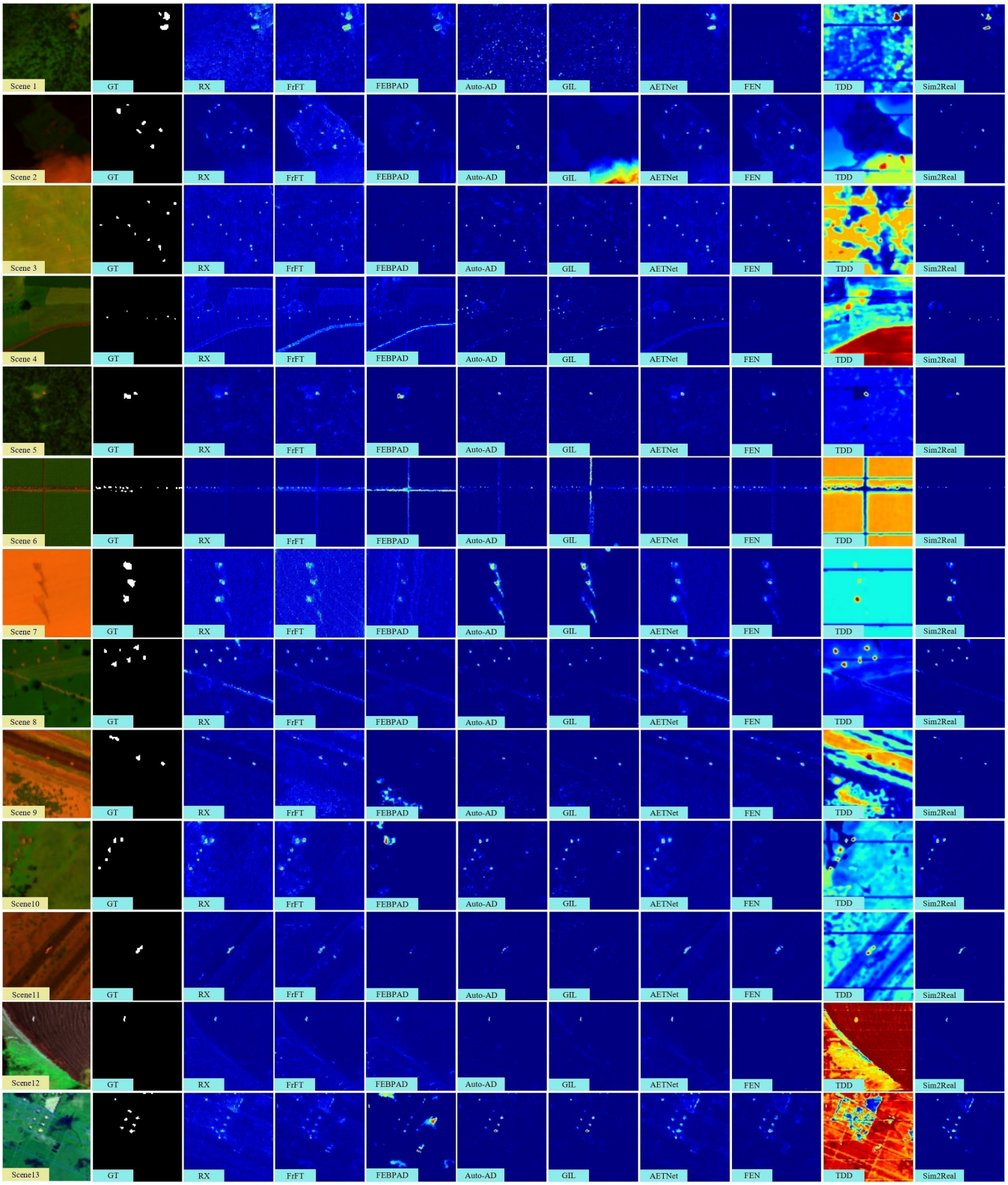


Fig. 9. Quantitative comparison results from some scenes in the HAD100 and ABU datasets.

2) *Without AM (Anomalous Modeling):* Similarly, variations in the spectral reflectance magnitude of anomalous targets were removed.
3) *Without NM (Noise Modeling):* Noise disturbances were removed.
4) *Without BS (Backgrounds Simulation):* The backgrounds of the simulated HADSim data were replaced with AVIRIS-NG backgrounds.

As shown in Table V, removing background, target, or noise interference leads to a performance drop, indicating the

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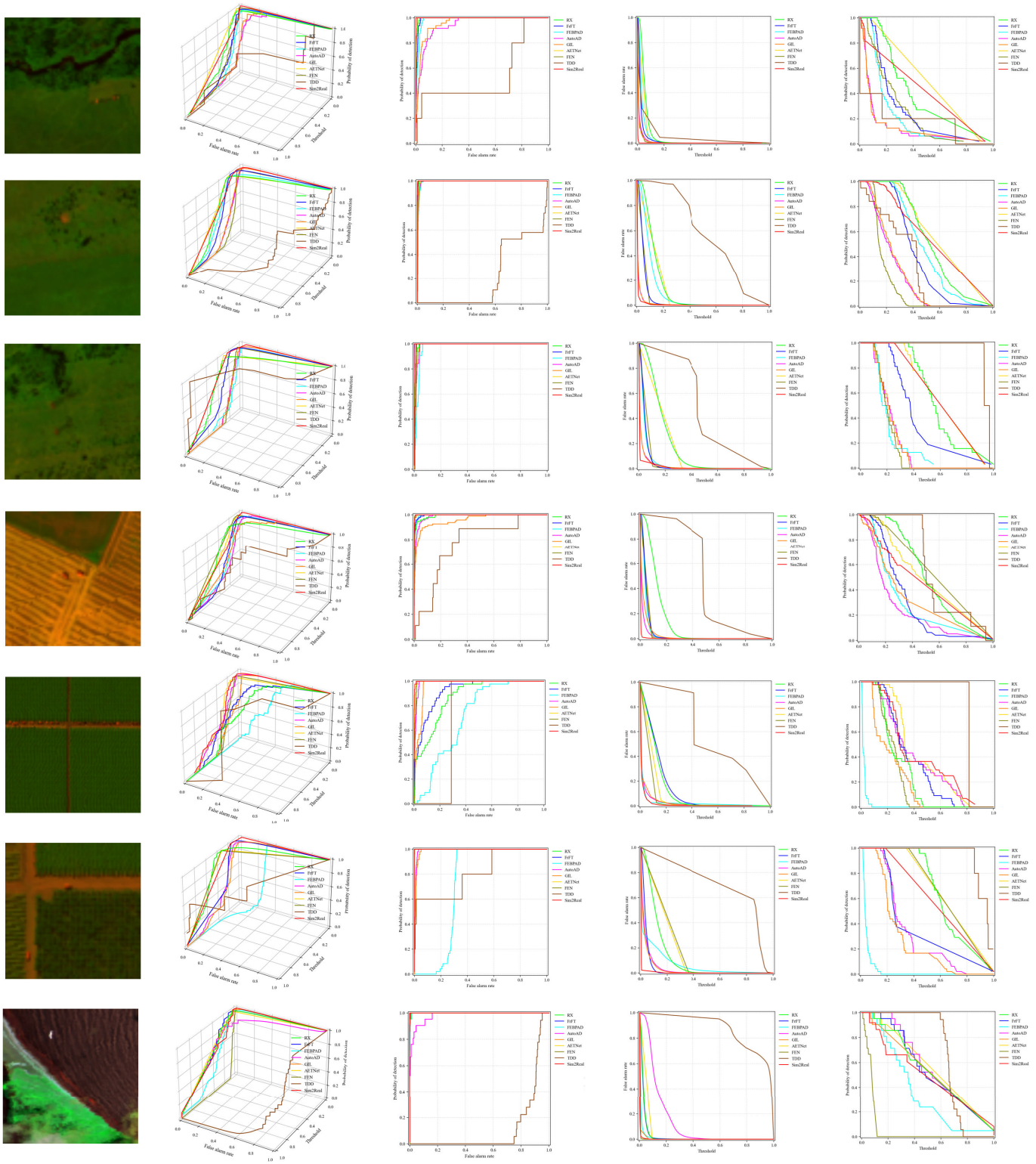


Fig. 10. Partial ROC comparison results from HAD100 and ABU dataset scenes. From (left to right) pseudo-color images, 3-D ROC, ROC curve of (PD, PF), ROC curve of (PF, threshold), and ROC curve of (PD, threshold).

effectiveness of the proposed simulation strategy. Additionally, replacing the simulated background with real backgrounds does not significantly improve detection performance, only slightly improving some metrics. This further confirms the validity of the proposed simulation strategy.

2) Effectiveness of Model Constraints and Structure: Experiments with different model configurations were conducted to verify the effectiveness of the proposed network.

1) *Without RXC:* The RXC module was removed from the Sim2Real network.

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TABLE V

ABLATION STUDY: QUANTITATIVE RESULTS COMPARISON OF DIFFERENT HADSIM VARIANTS

Variants	$AUC_{(D,F)}$	$AUC_{(F,\tau)}$	AUC_{BS}	AUC_{SNPR}	AUC_{OADP}
wo. BM	0.9876	0.0095	0.9781	106.99	2.1490
wo. AM	0.9863	0.0062	0.9801	123.38	2.2153
wo. NM	0.9868	0.0083	0.9785	110.74	2.2184
wo. BS	**0.9896**	0.0056	0.9840	126.30	**2.3225**
Sim2Real	0.9893	**0.0048**	**0.9846**	**131.12**	2.3184

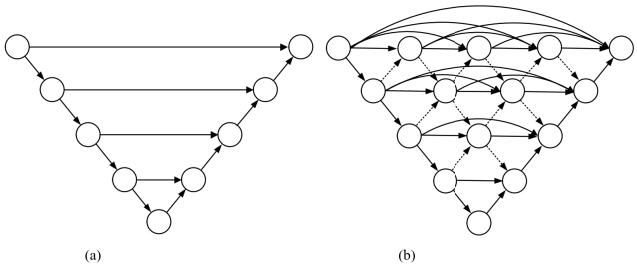


Fig. 11. Two Sim2Real variants, with (a) and (b) two different encoder–decoder backbones, respectively.

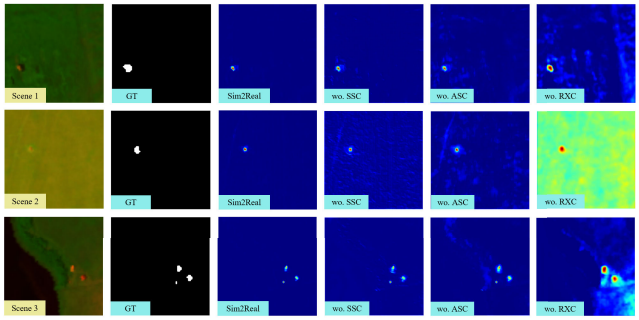


Fig. 12. AD results obtained by Sim2Real and its three variants.

2) *Without ASC:* The ASC module was removed.
3) *Without SSC:* The SSC module was removed.
4) *Without DC (Dense Connections):* As shown in Fig. 11(a), the dense connections at the original scale were removed, using U-Net as a baseline for comparison.
5) *With DNC (Dense Nested Connections):* As shown in Fig. 11(b), we introduced additional dense nested connections into Sim2Real, inspired by DNA-Net [66], to construct a comparative variant.

Table VI summarizes the experimental metrics for these variants. Removing any of the three constraints leads to performance degradation, demonstrating their effectiveness. Removing RXC causes a significant drop, especially in the AUC_{SNPR} metric, showing that using RX global spectral constraints as a prior helps reduce false alarms from background. Fig. 12 further visualizes anomaly maps under different variants, confirming the effect of RXC. Both SSC and ASC constraints also reduce the probability of false alarms from the background, further validating the proposed strategies.

Furthermore, for different structural variants, removing dense connections at the original scale leads to a clear degradation in all metrics, confirming the effectiveness of dense skip connections. Replacing the backbone with a more densely connected network significantly increases the parameter count, but the model performance does not improve noticeably, indicating that the current Sim2Real backbone achieves a favorable balance between efficiency and performance.

TABLE VI

ABLATION STUDY: QUANTITATIVE RESULTS COMPARISON OF DIFFERENT SIM2REAL VARIANTS

Variants	Params	$AUC_{(D,F)}$	$AUC_{(F,\tau)}$	AUC_{BS}	AUC_{SNPR}	AUC_{OADP}
wo. RXC	0.77M	0.9186	0.1447	0.7739	18.337	2.1381
wo. ASC	0.77M	0.9815	0.0108	0.9707	100.97	**2.3266**
wo. SSC	0.77M	0.9868	0.0086	0.9782	118.80	2.3125
wo. DC	0.49M	0.9853	0.0069	0.9784	125.18	2.2587
w. DNC	1.21M	0.9889	**0.0046**	0.9843	**136.60**	2.3213
Sim2Real	0.77M	**0.9893**	0.0048	**0.9846**	131.12	2.3184

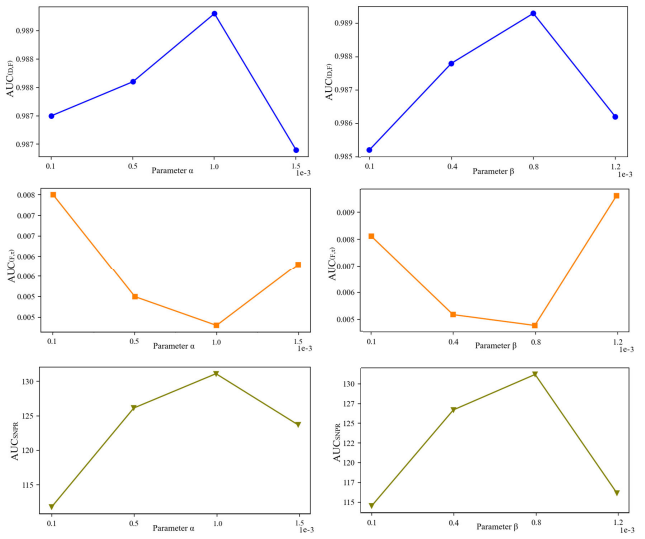


Fig. 13. Hyperparameter analysis of loss function weights: (left to right) parameters α and β; (top to bottom) $AUC_{(D,F)}$, $AUC_{(F,\tau)}$, and AUC_{SNPR}.

TABLE VII

ABLATION STUDY: QUANTITATIVE RESULTS COMPARISON OF DIFFERENT SIM2REAL VARIANTS

Variants	Params	$AUC_{(D,F)}$	$AUC_{(F,\tau)}$	AUC_{BS}	AUC_{SNPR}	AUC_{OADP}
32-16-16-8-8	0.22M	0.9618	0.0183	0.9435	112.57	2.1071
32-32-16-8-8	0.30M	0.9807	0.0089	0.9718	116.44	2.2543
64-32-16-8-8	0.77M	**0.9893**	**0.0048**	**0.9846**	131.12	2.3184
128-64-32-16-8	3.00M	0.9888	0.0061	0.9827	**136.09**	**2.3250**
256-128-64-32-16	11.8M	0.9785	0.0105	0.9680	88.755	2.1883

3) Parameter Sensitivity and Settings: In the proposed model, the weights of ASC and SSC losses are critical parameters. Fig. 13 shows the variation of three key metrics with respect to these parameters, indicating that ASC and SSC achieve optimal results when set to $1e^{-3}$ and $8e^{-4}$, respectively. Furthermore, experiments on different channel

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numbers (from shallow to deep U-Net stages) were conducted. As shown in Table VII, moderately increasing the number of channels improves HAD detection performance. However, when the parameter scale exceeds 64-32-16-8-8, performance improvements tend to saturate and some metrics decline; further increasing the model parameters may lead to overfitting. Therefore, 64-32-16-8-8 is a reasonable channel configuration. In addition, based on empirical observations, the ASC downsampling range is set to 11, the background disturbance parameter range to 1.0–2.0, and the target disturbance parameter range to 0.6–1.5.

V. Conclusion

This article presents an HAD study for HSI and proposes the image simulation-driven end-to-end AD framework Sim2Real-HAD. First, an HSI simulation method for AD is proposed, which models spatial and spectral variations of anomalies and background, providing rich and sufficient training data for AD models. Second, based on the simulated data, an end-to-end HAD network Sim2Real is developed, incorporating RX statistical prior constraints, SSCs, and spatial sparsity constraints for anomalous targets. By addressing the data scarcity problem with simulated data, the model enables direct mapping from HSI to anomaly results in an end-to-end learning manner. The three constraints further improve detection performance. Finally, extensive experiments comparing Sim2Real-HAD with state-of-the-art methods on real scenes demonstrate its superiority. Moreover, ablation studies confirm the effectiveness of the proposed improvements.

In future work, we will further investigate explicit spectral variability modeling, such as band-correlated perturbations and nonlinear spectral shifts, to improve the realism and diversity of simulated hyperspectral backgrounds and enhance adaptability to complex real-world scenes.

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Kewen Huang, photograph and biography not available at the time of publication.

Jie Wang, photograph and biography not available at the time of publication.

Ke Li (Graduate Student Member, IEEE), photograph and biography not available at the time of publication.

Yiping Xu, photograph and biography not available at the time of publication.



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